

An Internship Report
Of
C++ Programming

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B-Tech (CSE)



By

Roll Number of Student:- 25SCS1003005486

Name of Student: Khushi Jaiswal

Batch: 2026-27

CANDIDATE'S DECLARATION

I, **KHUSHI JAISWAL**, do hereby declare that the internship report titled “C++ Programming” has been completed by me to fulfill the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been duly quoted, and I have not taken any material as such from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement. I further declare that I shall be solely responsible for any kind of copyright violation in this regard.

Student Name: **KHUSHI JAISWAL**

Roll No.: **25SCS1003005486**

ACKNOWLEDGEMENT

I am using this opportunity to express my sincere gratitude to everyone who supported and guided me throughout my internship program. I am thankful for their valuable guidance, constructive feedback, and continuous encouragement during the internship. Their support helped me gain practical knowledge and improve my understanding of C++ Programming.

I would like to express my sincere gratitude to **Thiranex – Skill Development & Future Tech** for providing me with the opportunity to undertake an internship in C++ Programming. This internship provided me with valuable exposure to programming concepts, practical project-based learning, and an opportunity to enhance my technical skills. I am grateful to the mentors and team members at **Thiranex** for their guidance and support throughout the internship. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support, encouragement, and guidance.

Finally, I express my heartfelt gratitude to my parents for their constant support, encouragement, and blessings.

Signature:

Student Name: **KHUSHI JAISWAL**

Roll No.: **25SCS1003005486**

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3. INTERNSHIP COMPLETION CERTIFICATE



Internship Offer Letter



Date: 30 June 2026

TO,

Khushi Jaiswal

Intern ID: THX-JUN3026-435

Subject: Offer Letter for Internship in C++ Programming

Dear **Khushi Jaiswal**,

Following our selection process, we are pleased to offer you an internship position at **Thiranex**. We believe your skills and enthusiasm will be a great addition to our team.

Internship Role	Intern - C++ Programming
Commencement Date	30 Jun 2026
Completion Date	29 Jul 2026
Work Mode	Remote / Project-Based

During this tenure, you will work on practical projects under industry mentorship. Your progress will be reviewed periodically, and your performance will determine the successful completion of the internship.

We look forward to a mutually beneficial learning journey. Please confirm your acceptance by starting your onboarding tasks.

This is an electronically generated document. No physical signature is required.
For verification, contact Thiranex Verification Cell.

www.thiranex.in



WEEK-WISE STRUCTURE OF THE INTERNSHIP

The week-wise structure of the internship was organized around practical C++ programming projects. Each project focused on applying programming concepts such as object-oriented programming, file handling, data management, conditional logic, loops, arrays, and menu-driven operations.

Week	Project / Module	Description
Week 1	Student Management System	Developed a console-based Student Management System in C++. Implemented features to add, update, delete, and display student information. File handling was used for persistent storage of student records.
Week 2	Bank Management Application	Developed a C++ based Bank Management Application using object-oriented programming and file handling. Implemented core banking operations such as deposit, withdrawal, and balance checking while maintaining customer records.
Week 3	Library Management System	Developed a Library Management System to manage books, members, and borrowing records. Implemented book issue and return functionality along with searching books by title or author.
Week 4	Mini Game Project – Tic Tac Toe / Snake Game	Created a console-based mini game using C++. Applied loops, arrays, conditional statements, and game logic. Implemented dynamic board display, win/loss detection, and replay functionality.

Overall Internship Outcome

The internship provided practical experience in C++ programming and helped in understanding the implementation of programming concepts through real-world project-based tasks. The projects improved skills in object-oriented programming, file handling, data management, problem-solving, and logical thinking.

4. INTERNSHIP PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the internship undertaken at **Thiranex** in the domain of **C++ Programming**. The internship was carried out from **30 June 2026 to 29 July 2026** as part of the practical learning requirements of the Bachelor of Technology (Computer Science and Engineering) programme.

The primary aim of this internship was to strengthen programming skills by applying theoretical concepts of C++ to practical, project-based applications. During the internship, the learner gained hands-on experience in important programming concepts such as **Object-Oriented Programming (OOP), file handling, data management, loops, arrays, conditional statements, and menu-driven programming**.

Throughout the internship, several practical projects were completed, including a **Student Management System, Bank Management Application, Library Management System, and Mini Game Project (Tic Tac Toe / Snake Game)**. These projects provided practical exposure to designing console-based applications, managing records, implementing different functionalities, and developing logical problem-solving skills.

The internship followed a project-based learning approach in which each assigned task focused on implementing specific C++ programming concepts. Successful completion of the projects helped in developing a better understanding of software development, programming logic, data persistence through file handling, and the practical application of C++ concepts.

Overall, the internship provided valuable practical experience in C++ programming and helped bridge the gap between academic knowledge and real-world application development.

4.2 Organization Profile

Thiranex is a skill development and future technology organization focused on providing practical, industry-oriented learning opportunities to students and aspiring professionals. The organization aims to bridge the gap between academic knowledge and practical skills by providing project-based training and internship opportunities in various technology domains.

Thiranex provides learners with opportunities to gain hands-on experience through practical assignments and real-world projects. Its internship programs are designed to help students strengthen their technical knowledge, improve problem-solving abilities, and develop skills that are relevant to the software and technology industry.

As part of the **C++ Programming Internship**, the training focused on applying fundamental and advanced C++ programming concepts through practical projects. The internship provided exposure to **Object-Oriented Programming, file handling, data management, loops, arrays, conditional statements, and menu-driven programming.**

During the internship, various console-based applications were developed, including a **Student Management System, Bank Management Application, Library Management System, and Mini Game Project.** These projects helped in developing programming logic and understanding how C++ concepts can be applied to create functional software applications.

Overall, Thiranex provided a practical learning environment that enabled the learner to enhance programming skills, gain project development experience, and understand the practical application of C++ in software development.

4.3 Problem Statement

Although many engineering students learn programming concepts as part of their academic coursework, they often have limited opportunities to apply these concepts in practical software development projects. Understanding the syntax of a programming language is only one part of software development; applying programming logic to solve real-world problems and developing functional applications is equally important.

The problem addressed during this internship was to gain practical experience in **C++ programming** by implementing various concepts through project-based applications. The internship focused on understanding and applying concepts such as **Object-Oriented Programming, file handling, data management, loops, arrays, conditional statements, and menu-driven programming.**

To address these practical requirements, multiple console-based projects were developed, including a **Student Management System, Bank Management Application, Library Management System, and Mini Game Project (Tic Tac Toe / Snake Game).** These projects involved managing records, performing different operations, storing data using files, implementing application logic, and providing interactive menu-driven functionality.

The internship therefore provided an opportunity to bridge the gap between theoretical programming knowledge and practical application development by designing, implementing, testing, and improving functional C++ applications.

4.4 Project Objectives

The major objectives of the C++ Programming Internship were:

- To develop a strong understanding of the fundamentals of **C++ programming** and its practical applications.
- To understand and apply **Object-Oriented Programming (OOP)** concepts such as classes, objects, encapsulation, inheritance, and polymorphism.
- To learn and implement **file handling** techniques for storing, retrieving, and managing data efficiently.
- To improve programming logic and problem-solving skills through practical, project-based assignments.
- To develop interactive **menu-driven console applications** using C++.
- To gain practical experience in developing applications for managing **student, banking, and library records**.
- To understand the implementation of operations such as adding, updating, deleting, searching, issuing, returning, depositing, and withdrawing data in C++ applications.
- To apply programming concepts such as **loops, arrays, conditional statements, functions, and data structures** while developing projects.
- To develop a **mini game project** using C++ programming logic and improve understanding of game-based application development.
- To test, debug, and improve C++ programs while ensuring that the developed applications perform their intended functions correctly.
- To bridge the gap between theoretical knowledge gained during academic studies and its **practical implementation in software development**.
- To gain hands-on experience in developing complete, functional C++ projects from planning and implementation to testing and completion.

4.5 Scope of the Internship Projects

The scope of this internship focused on the practical application of **C++ programming concepts** through the development of multiple console-based projects. The internship covered important programming areas such as Object-Oriented Programming, file handling, data management, functions, loops, arrays, conditional statements, and menu-driven programming.

The projects developed during the internship included a **Student Management System, Bank Management Application, Library Management System, and Mini Game Project (Tic Tac Toe / Snake Game)**. These applications involved managing student and customer records, performing banking operations, managing books and borrowing records, and implementing interactive game logic.

The scope also included the use of **file handling for persistent data storage**, allowing information to be stored and retrieved when required. The projects emphasized logical problem-solving, application design, implementation, testing, and debugging using C++.

The internship was primarily **project-based and focused on practical learning**. The developed applications were console-based and intended for educational and skill-development purposes rather than deployment as large-scale commercial systems. Through these projects, the internship provided hands-on experience in applying C++ programming concepts to functional software applications.

4.6 Technologies and Tools Used

The following technologies and tools were used during the development of the projects undertaken as part of the C++ Programming Internship:

- **Programming Language:** C++
- **Programming Concepts:** Object-Oriented Programming (OOP), functions, loops, arrays, conditional statements, and data structures.
- **File Handling:** C++ file handling techniques for storing, retrieving, updating, and managing records persistently.
- **Development Environment:** C++ compiler and an Integrated Development Environment (IDE) for writing, compiling, testing, and debugging programs.

- **Application Development:** Console-based, menu-driven application development using C++.
- **Data Management:** File-based storage for managing student, customer, book, and game-related data.
- **Project Development:** Practical implementation of C++ concepts through Student Management System, Bank Management Application, Library Management System, and Mini Game Project.
- **Testing and Debugging:** Compilation, execution, error identification, debugging, and functional testing of the developed C++ applications.

4.7 System Architecture

The general architecture followed during the internship focused on the development of **console-based C++ applications**. The projects were designed using a simple and structured approach in which user input was processed through a menu-driven interface, the required operations were performed using C++ programming logic, and data was stored using file handling wherever required.

The architecture of the developed applications can be summarized through the following layers:

- **User Interface Layer:** Provides a menu-driven console interface through which users can select different operations and enter the required information.
- **Input and Validation Layer:** Accepts user input and performs basic validation to ensure that the entered information is appropriate for the selected operation.
- **Application Logic Layer:** Contains the main C++ programming logic, including functions, loops, conditional statements, arrays, and other programming constructs required to perform operations.
- **Object-Oriented Programming Layer:** Classes and objects are used where required to organize data and functionality, particularly in the Bank Management and Library Management applications.
- **File Handling Layer:** C++ file handling is used to store and retrieve records persistently. This layer manages data related to students, customers, books, and other application records.
- **Data Management Layer:** Handles operations such as adding, updating, deleting, searching, displaying, issuing, returning, depositing, and withdrawing information according to the requirements of each project.

- **Output Layer:** Displays processed information, transaction results, records, game boards, and appropriate messages to the user through the console.

The overall architecture provides a structured approach to developing functional C++ applications while maintaining clear separation between user interaction, application logic, data processing, and persistent storage.

4.8 Methodology

The internship followed a structured, project-based methodology combining conceptual understanding, practical implementation, testing, debugging, and project completion. Each assigned project focused on applying specific C++ programming concepts to develop functional console-based applications.

The methodology followed during the internship is summarized below:

Week	Project / Module	Description
Week 1	Student Management System	Developed a console-based application to manage student records. Implemented functionalities such as adding, updating, deleting, and displaying student information. File handling was used for persistent data storage.
Week 2	Bank Management Application	Developed a banking application using C++ and Object-Oriented Programming concepts. Implemented deposit, withdrawal, and balance-checking operations along with customer record management using file handling.
Week 3	Library Management System	Developed an application for managing books, members, and borrowing records. Implemented book addition, issue and return operations, and search functionality based on title or author.
Week 4	Mini Game Project – Tic Tac Toe / Snake Game	Developed a console-based mini game using loops, arrays, conditional statements, and logical programming concepts. Implemented dynamic game display, win/loss detection, and replay functionality.

During the internship, each project was developed through different stages including **requirement understanding, program design, coding, compilation, testing, debugging, and final implementation**. The projects were completed according to the assigned requirements and helped in applying theoretical C++ concepts to practical applications.

The project-based methodology provided hands-on experience in developing functional programs and strengthened skills in **programming logic, problem-**

solving, Object-Oriented Programming, file handling, data management, and application.

4.9 Project Modules and Features

The internship projects were divided into different modules based on their specific requirements and functionalities. Each project focused on implementing practical C++ programming concepts and developing a functional console-based application.

4.9.1 Student Management System

Modules:

- Student Record Management
- Add Student Information
- Update Student Information
- Delete Student Records
- Display Student Records
- File-Based Data Storage

Key Features:

- Allows users to add new student information.
- Provides options to update existing student records.
- Allows deletion of student records when required.
- Displays stored student information in an organized manner.
- Uses file handling for persistent storage of student data.
- Provides a simple menu-driven console interface.

4.9.2 Bank Management Application

Modules:

- Customer Account Management
- Deposit Module
- Withdrawal Module
- Balance Inquiry
- Customer Data Storage

Key Features:

- Maintains customer and account information.
- Allows users to deposit money into an account.
- Provides withdrawal functionality.
- Allows users to check the available account balance.
- Uses Object-Oriented Programming concepts for organizing data and operations.
- Uses file handling for maintaining customer records.

4.9.3 Library Management System

Modules:

- Book Management
- Member Management
- Book Issue Module
- Book Return Module
- Book Search Module

Key Features:

- Allows addition and management of book details.
- Maintains member information.
- Provides functionality to issue books to members.
- Allows books to be returned after borrowing.
- Provides search functionality based on book title or author.
- Helps organize borrowing and library records efficiently.

4.9.4 Mini Game Project – Tic Tac Toe / Snake Game

Modules:

- Game Board Management
- Player Input
- Game Logic
- Win/Loss Detection
- Replay Functionality

Key Features:

- Provides an interactive console-based gaming experience.
- Uses arrays to represent and manage the game board.
- Uses loops and conditional statements for game logic.
- Dynamically displays the game board after each move.
- Detects winning, losing, or game-completion conditions.
- Provides an option to replay the game.

Summary of Project Features

Project	Main Features
Student Management System	Add, update, delete, display, and store student records
Bank Management Application	Deposit, withdrawal, balance check, and customer record management
Library Management System	Book addition, issue, return, member management, and search
Mini Game Project	Game board, player input, game logic, win/loss detection, and replay

These modules and features collectively provided practical experience in **C++ programming, file handling, Object-Oriented Programming, data management, conditional logic, loops, arrays, and menu-driven application development.**

4.10 Expected Outcomes

The expected outcomes of the C++ Programming Internship were:

- Strong understanding of **C++ programming fundamentals** and their practical implementation.
- Ability to develop functional **console-based applications** using C++ programming concepts.
- Practical understanding of **Object-Oriented Programming (OOP)** and its application in software development.

- Ability to implement **file handling** for persistent storage, retrieval, and management of application data.
- Improved programming logic and **problem-solving skills** through project-based development.
- Practical experience in developing and managing records through the **Student Management System, Bank Management Application, and Library Management System**.
- Ability to implement **menu-driven operations**, including adding, updating, deleting, displaying, searching, depositing, withdrawing, issuing, and returning records.
- Improved understanding of **loops, arrays, conditional statements, functions, and data management** through the development of the assigned projects.
- Practical experience in designing and implementing a **Mini Game Project (Tic Tac Toe / Snake Game)** using C++ programming logic.
- Improved skills in **program testing, debugging, error handling, and performance improvement**.
- Ability to apply theoretical programming knowledge to develop **functional and user-oriented software applications**.
- Successful completion of the assigned C++ projects, demonstrating practical knowledge and programming skills gained during the internship.

4.11 Project Implementation And Results

During the internship, four practical projects were implemented using **C++ programming**. Each project was developed to apply different programming concepts and improve practical problem-solving skills. The implementation process involved understanding the requirements, designing the program structure, writing the code, testing the functionalities, identifying errors, and making necessary corrections.

4.11.1 Student Management System

The Student Management System was developed as a console-based C++ application for managing student records. The system provided functionalities to **add, update, delete, and display student information**. File handling was implemented to store student data persistently so that records could be accessed when required.

Result: The project successfully provided a simple and organized way to manage student records and demonstrated the practical use of file handling and menu-driven programming.

4.11.2 Bank Management Application

The Bank Management Application was developed using **C++ and Object-Oriented Programming concepts**. The application implemented basic banking operations such as **deposit, withdrawal, and balance checking**. Customer information was managed using file handling for persistent record storage.

Result: The application successfully simulated basic banking operations and provided practical experience in applying classes, objects, functions, and file handling in C++.

4.11.3 Library Management System

The Library Management System was developed to manage **books, members, and borrowing records**. The application included features for adding books, issuing and returning books, and searching for books based on their title or author.

Result: The project successfully demonstrated how C++ programming can be used to organize library operations and manage book and member information efficiently.

4.11.4 Mini Game Project – Tic Tac Toe / Snake Game

A console-based Mini Game Project was developed using fundamental C++ programming concepts. The project applied **loops, arrays, conditional statements, and logical operations** to implement game functionality. Features such as dynamic board display, win/loss detection, and replay options were incorporated.

Result: The mini game was successfully implemented as an interactive console application and helped strengthen programming logic, conditional reasoning, and problem-solving skills.

Overall Results

All assigned projects were successfully completed during the internship. The implementation of these projects provided practical experience in **C++ programming, Object-Oriented Programming, file handling, data management, logical problem-solving, debugging, and console-based application development**. The projects successfully met their intended objectives and helped in applying academic programming knowledge to practical software development tasks.

5.CONCLUSIONS AND SUGGESTIONS

Conclusion

The C++ Programming Internship at **Thiranex** was a valuable learning experience that provided practical exposure to programming and application development. The internship helped in understanding how theoretical concepts of C++ can be applied to develop functional and user-oriented software applications.

During the internship, I successfully worked on four practical projects: **Student Management System, Bank Management Application, Library Management System, and Mini Game Project (Tic Tac Toe / Snake Game)**. These projects provided hands-on experience in implementing important concepts such as **Object-Oriented Programming, file handling, functions, loops, arrays, conditional statements, data management, and menu-driven programming**.

The development of these projects improved my programming logic, problem-solving ability, debugging skills, and understanding of software development. Working on different types of applications also helped me understand how programming concepts can be combined to solve practical problems and create useful applications.

Overall, the internship successfully bridged the gap between academic knowledge and practical implementation. It strengthened my confidence in C++ programming and provided a strong foundation for further learning and development in the field of software engineering.

Suggestions

Based on my internship experience, the following suggestions can be considered for further improvement:

- More advanced C++ projects can be introduced to provide deeper practical exposure.
- Students can be given opportunities to work with **database management systems** along with file handling.
- Advanced C++ concepts such as **STL, templates, exception handling, and advanced OOP** can be included in future projects.
- Projects with graphical user interfaces can be introduced to help students move beyond console-based applications.
- Students can be encouraged to work on larger projects involving multiple modules and real-world use cases.
- Regular code reviews and feedback sessions can help students improve their programming practices and coding standards.
- Version control tools such as **Git and GitHub** can be incorporated to provide experience with professional software development workflows.

The internship was an important step in developing my technical skills and practical understanding of C++ programming. The knowledge and experience gained during this internship will be useful for future academic projects as well as professional software development opportunities.

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